Week 02: Tables, keys and constraints

Learning outcomes: CLO1, CLO2, CLO3

Lesson

- **Schema:** the table structure (columns, types, keys). **Instance:** the rows stored right now.
- Every column has a **data type**. P-codes are text (`CHAR`); areas are `DECIMAL(12,1)` (exact, unlike `FLOAT`).
- **Candidate key:** a minimal set of columns that identifies each row. The one you choose is the **primary key** (unique, not NULL).
- **Foreign key:** value must exist in the parent table. **NOT NULL:** value required. **CHECK:** value must pass a test, e.g. `area_km2 > 0`.
- `cambodia` is read-only for you. Create your tables in `student_work`.
- To see a constraint work, insert a row that already exists and save the error.

Worked example

Run `example.sql` (how (`setup/README.md`)), then answer in `work.sql`.

Tables for this week

Read `cambodia`, write `student_work`. All columns are `NOT NULL`. Samples are the first two rows by code.

`cambodia.province`

Column	Type	Key	Meaning
`province_pcode`	`CHAR(4)`	PK	Province code
`province_name`	`VARCHAR(100)`		Province name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place

Sample rows:

province_pcode	province_name	area_km2
KH01	Banteay Meanchey	6145.9
KH02	Battambang	11869.0

`cambodia.district`

Column	Type	Key	Meaning
`district_pcode`	`CHAR(6)`	PK	District code
`district_name`	`VARCHAR(100)`		District name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`province_pcode`	`CHAR(4)`	FK	Province code

Sample rows:

district_pcode	district_name	area_km2	province_pcode
KH0102	Mongkol Borei	807.2	KH01
KH0103	Phnum Srok	761.7	KH01

`cambodia.commune`

Column	Type	Key	Meaning
`commune_pcode`	`CHAR(8)`	PK	Commune code
`commune_name`	`VARCHAR(100)`		Commune name (English)
`area_km2`	`DECIMAL(12,1)`		Area in km², one decimal place
`district_pcode`	`CHAR(6)`	FK	District code

Sample rows:

commune_pcode	commune_name	area_km2	district_pcode
KH010201	Banteay Neang	52.0	KH0102
KH010202	Bat Trang	40.3	KH0102

Join keys: `district.province_pcode = province.province_pcode`; `commune.district_pcode = district.district_pcode`.

Your tables: `example.sql` creates `student_work.w02_province`. You create the district and commune tables in the lab.

More: data guide (`data/README.md`).

Lab tasks

1. Explain the data type of each column in the three tables.
2. Create `student_work.lab02_province` with a PK, NOT NULL columns and `CHECK (area_km2 > 0)`.
3. Copy the first two provinces by code into it with `INSERT ... SELECT`.
4. Insert one of them again and save the duplicate-key error.
5. Create `student_work.lab02_district` with an FK to `lab02_province` and load only matching districts. Why load parents first?

Homework

1. Create `student_work.lab02_commune` with an FK to `lab02_district` and load only matching communes.
2. Run `SHOW CREATE TABLE` and say what each constraint blocks.
3. Add a nullable column to `lab02_province` with ALTER TABLE. Then delete a province that still has districts in `lab02_district` and save the FK error. Explain ON DELETE RESTRICT vs CASCADE.

Capstone sprint

C2: Draft schema v0 (tables, columns, types, keys) and a data dictionary.

Submit

Answers in `work.sql`, reflection in `reflection.md`, output in `results/`. Commit and push. See SUBMISSIONS.md.